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INSTITUTO
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Dissertação em Engenharia

Research Support Tools

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Outline

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1. Information Sources

- Search for information is required at various stages of a thesis project.
- Reliable references should be used for strengthening your arguments in the thesis document.
- By simply looking at the reference list, an experienced reviewer can see if sources of good quality were used.
- Depending on the research question and time frame (limited in M.Sc theses), it may not be viable to apply a method within the 'systematic literature review' family
 - an ad-hoc/narrative literature review can be used instead

2. Types of Information Sources

- **Theses (M.Sc., Ph.D.)**

- can contain valuable information with the state of the art for a subject.
- they are usually of adequate quality as theses are examined and assessed.
- typically a Ph.D. thesis has more material than a M.Sc thesis.

- **Scientific research journals**

- typically publish high quality articles.
- written by experts in the field for other experts (difficult to understand for non-experts).
- important to check if research findings may have become somewhat outdated.
- Important to check the quality of the journal
- time for publication can be long, except for letters and Open Access journals

2. Types of Information Sources

- **Conference and workshop proceedings**
 - publish quality papers relatively fast (compared to traditional journals)
 - Review process is often less demanding than for journals → important to check reputation.
- **Books, Book chapters**
 - usually a good source for understanding the fundamentals in a variety of subjects and areas.
 - often do not cover the latest research findings.
- **Magazines**
 - popular version of research journals.
 - can contain articles on research results with simplifications, to make the material more accessible to non-experts.

2. Types of Information Sources

- **Web pages**

- can contain useful information, but in general, they should be avoided since their content is normally not reviewed.
- they can also disappear or change.

- **Newspapers**

- can provide interesting pieces of information, which can be used as examples.
- typically lack most of the details and are not reviewed by an expert in the field.

- **Other documents**

- white papers, manuals, modelling documents, and commercial information.
- should also be treated with care since there is no guarantee that they are of good quality.

3. Information seeking

- After formulating the search problem and what information you need, it is time to choose your search tools.
- Article databases
 - IEEE Xplore, <https://ieeexplore.ieee.org/>
 - ACM Digital Library, <https://dl.acm.org/>
 - ScienceDirect, <https://www.sciencedirect.com/>
 - SpringerLink, <https://link.springer.com/>
 - Wiley Online Library, <https://onlinelibrary.wiley.com/>
 - JSTOR, <https://www.jstor.org/>
 - SciELO, <https://scielo.org/>

3. Information seeking

- **Search services/discovery tools/aggregators**

- Google Scholar, <https://scholar.google.com>
- Scopus, <https://www.scopus.com/>
- Web of Science, <https://www.webofknowledge.com/>
- B-ON, <https://www.b-on.pt/>
- DBLP, <https://dblp.org/>
- Semantic Scholar, <https://www.semanticscholar.org/>
- Crossref, <https://search.crossref.org/>
- ResearchGate, <https://www.researchgate.net/>
- CORE (Open Access papers), <https://core.ac.uk/>
- Internet Archive Scholar (Open Access papers), <https://scholar.archive.org/>
- Directory of Open Access Journals: <https://doaj.org/>
- CiteSeerX (Computer and information science), <https://citeseerx.ist.psu.edu/>

Additional Links:

https://en.wikipedia.org/wiki/List_of_academic_databases_and_search_engines

3. Information seeking

- **The process of information searching can be performed in different ways:**
 - Selection of search terms and using an article database or search engine
 - Browsing
 - Special edition of a journal/conference in a specific topic
 - Publications of a known author or research group
 - (into the past) following references in one document.
 - (into the future) following citing references (using Google Scholar, Scopus, etc)

3. Information seeking

- Ferramentas IA

- Gemini,

- https://www.youtube.com/watch?v=sPiOP_CB54A&t=3s&ab_channel=Google

- Scopus AI, <https://www.scopus.com/search/form.uri?display=basic#scopus-ai>

4. Open Access Journals



Source: https://guides.library.ju.se/publishing-strategy/open_access

- There is a growing number of Open Access journals:
 - IEEE Access, <https://ieeaccess.ieee.org/>
 - MDPI journals, <https://www.mdpi.com/>
 - Hindawi journals, <https://www.hindawi.com/>
 - Proceedings of the ACM on Programming Languages, <https://dl.acm.org/journal/pacmpl>
 - Journal of Big Data, <https://journalofbigdata.springeropen.com/>
- Care must be taken regarding the quality of the publications (predatory publishers)

5. Preprints

- Full draft research papers can be shared publicly as preprints before peer review.
- Advantages:
 - Increase visibility
 - Receive early feedback
 - Get credit for novel contributions
- Can be cited in other research papers
- Care must also be taken regarding the quality when resorting to preprints.
- Examples of preprint repositories:
 - <https://arxiv.org/>
 - <https://www.techrxiv.org/>
 - <https://hal.archives-ouvertes.fr/>

6. Evaluation of sources

- **Many available sources of information → assess the credibility!**
- **Important to consider:**
 - Type of publication (Journal, conference or other)
 - Publisher
 - Authors and affiliations
 - Publication Date
 - References
 - Purpose and intended audience

7. Publications evaluation

- The scientific impact of an article is considered related to the number of citations it receives in other publications
- Citation databases provide information about how articles refer to a specific author or another publication
 - can also be used to measure the impact of an article
 - Examples : Web of Science, Google Scholar, Academic Search Complete, ScienceDirect, and [Scopus](#)
- The quality of a scholarly journal or a conference can be evaluated through its scientific influence
 - can be done through rankings, which can rely on metrics such as Impact factor/CiteScore, SCImago Journal Rank indicator, Eigenfactor score, h-index , etc.

7. Publications evaluation

- **Journal ranks can be consulted for example in:**
 - SCImago Journal Rank (SJR), <https://www.scimagojr.com/>
 - Web of Science, <https://jcr.clarivate.com/jcr/browse-categories>
 - Scopus, <https://www.scopus.com/sources>
 - Google Scholar,
https://scholar.google.com/citations?view_op=top_venues&hl=en
- **Rankings also exist for conferences:**
 - <http://www.conferenceranks.com/>
 - <https://www.scimagojr.com/>

8. Authors metrics

- The scientific impact of an individual author can also be evaluated using metrics, such as:
 - total citations
 - *h*-index
 - i10-index
- Several platforms enable a direct overview of an individual author's list of publications and metrics :
 - <https://scholar.google.com>
 - <https://orcid.org/>
 - <https://www.researchgate.net/>

Source: Berndtsson, M., Hansson, J., Olsson, B., Lundell, B.:
Thesis Projects: a Guide for Students in Computer Science
and Information Systems. Springer, London (2008)